Tresos Plug-in for Integration in DaVinci

Version 1.1.6
2023-02-20
Application Note AN-ISC-8-1241

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Restrictions Customer Confidential – for Tresos-Projects only

Abstract This application note mainly describes how to integrate a Tresos module in DaVinci Configurator Classic.

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1 Integration in DaVinci Configurator Classic

To integrate Tresos plug-ins into DaVinci Configurator Classic, follow these steps.



Note
The delivered SIP has one of two possible structures. In the following, one is referred to as component based (Figure 1-1), the other as legacy (Figure 1-2).

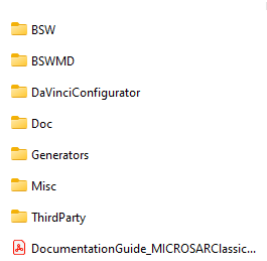


Figure 1-1 Component based structure

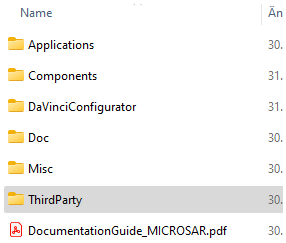


Figure 1-2 Legacy structure

1.1 Step 1: Install Tresos Plug-in

> Install the Tresos plug-in to any temporary folder (e.g. `Tmp\<NewModule>\`). After the installation you will find a subfolder in this folder named “plugins”. Only the content of this folder is relevant further on. Other files and folders can be ignored.

This step is only necessary if Tresos plug-ins are not installed already. This can be the case if further MCAL modules shall be integrated which are not part of the MCAL integration package included in the BSW Package (e.g., UART) or CDD modules.

If the MCAL or third party module vendor does not use Tresos for configuration and code generation but provides standalone executables, then use “UserManual_3rdParty-MCAL-Integration.pdf” for the integration.

If your BSW Package does not include an MCAL integration package, proceed with Step 3 (chapter 1.3).

Otherwise, when your BSW Package contains an MCAL integration package and when Tresos is used for generation, you can proceed with Step 2 (chapter 1.2).

1.2 Step 2: Integration of CDD modules

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Note
The integration helper (`3rdPartyMcalIntegrationHelper.exe`) supports the integration of CDD modules into the current BSW Package if it is technically useful. Please note, that this is **not possible for all platforms**. In latter case the CDD integration feature is not enabled. Furthermore, the feature is explicitly released for ASR 4 MCALs. If the CDD feature is not available, proceed with chapter 1.2.2.

> You can check, if the CDD feature is enabled for your platform by starting the integration helper (3rdPartyMcalIntegrationHelper.exe) and check if the button “Add / Remove CDD” exists:

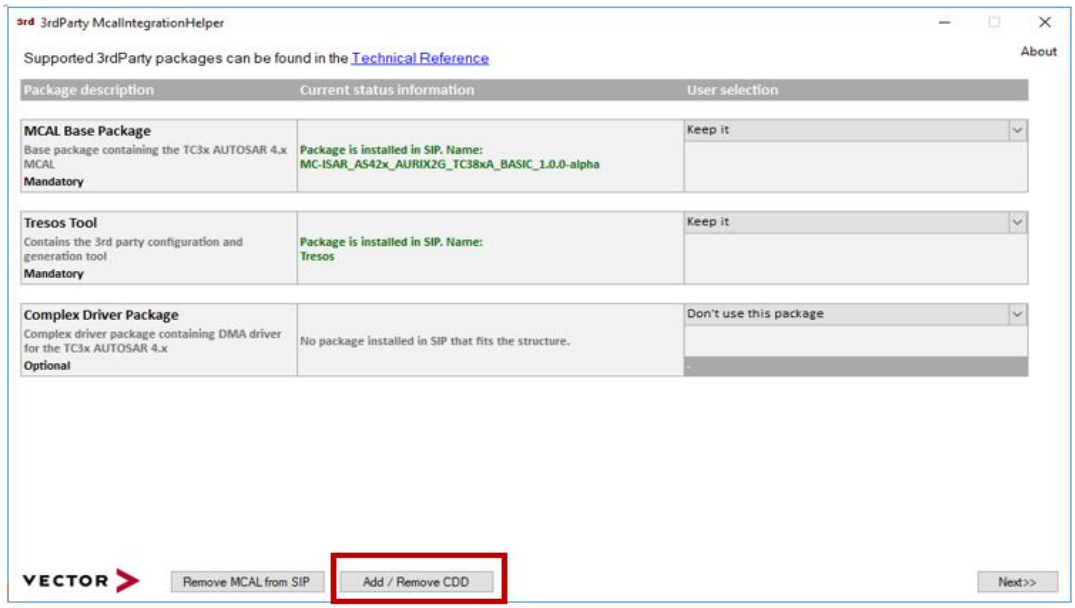


Figure 1-3 Add / Remove CDD button in the integration helper

> If the button “Add / Remove CDD” exists, you can continue with next chapter 1.2.1.
> If the button “Add / Remove CDD” does not exist, you can continue with chapter 1.2.2.

1.2.1 Step 2: Use of CDD integration feature

When the tool provides the CDD integration it performs the following tasks:

> Copy Tresos plug-in of a CDD module `e.g. Tmp\<NewModule>\plugins\NewModule_TS_TxxxxxRxx` into the “plugins” folder of the Tresos instance included in the BSW Package folder `ThirdParty\Mcal_xxx\Supply\Tresos\plugins`
> Create recommend configuration file for the CDD module
> Copy BSWMD file of CDD module into the BSW Package
> Create Tresos settings file that includes all generation parameters to start generation of `<NewModule>` in DaVinci Configurator Classic
> Create xsl transformation needed for generation of `<NewModule>`. As Tresos does not work with EcuC files, the module configuration must be transformed from EcuC to the Tresos format at first. That is why a transformation file is needed which strips down the overall EcuC file from DaVinci Configurator Classic to the relevant parts for the Tresos generator.

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Note
The MCAL is typically not included in the BSW Package after delivery. The customer has to copy his licensed MCAL to the empty third party folder at first by running the integration helper (`3rdPartyMcalIntegrationHelper.exe`).

Reference
Please refer to **UserManual_3rdParty-MCAL-Integration.pdf** for more information about the MCAL integration package provided with the MICROSAR Classic BSW Packages.

Please note that if your CDD module Tresos plug-in requires a different Tresos version than the one in the BSW Package, then it is not possible to use this CDD integration feature. The background is that it will integrate your CDD module using the Tresos version from BSW Package. Therefore, please continue with chapter 1.2.2, which describes how CDD module integration can be done without using this feature.

Otherwise, you can follow with next chapter.

1.2.1.1 Add-/Remove CDD modules

To start the feature, the integration helper (`3rdPartyMcalIntegrationHelper.exe`) must first be started and then the button “Add / Remove CDD” must be clicked:

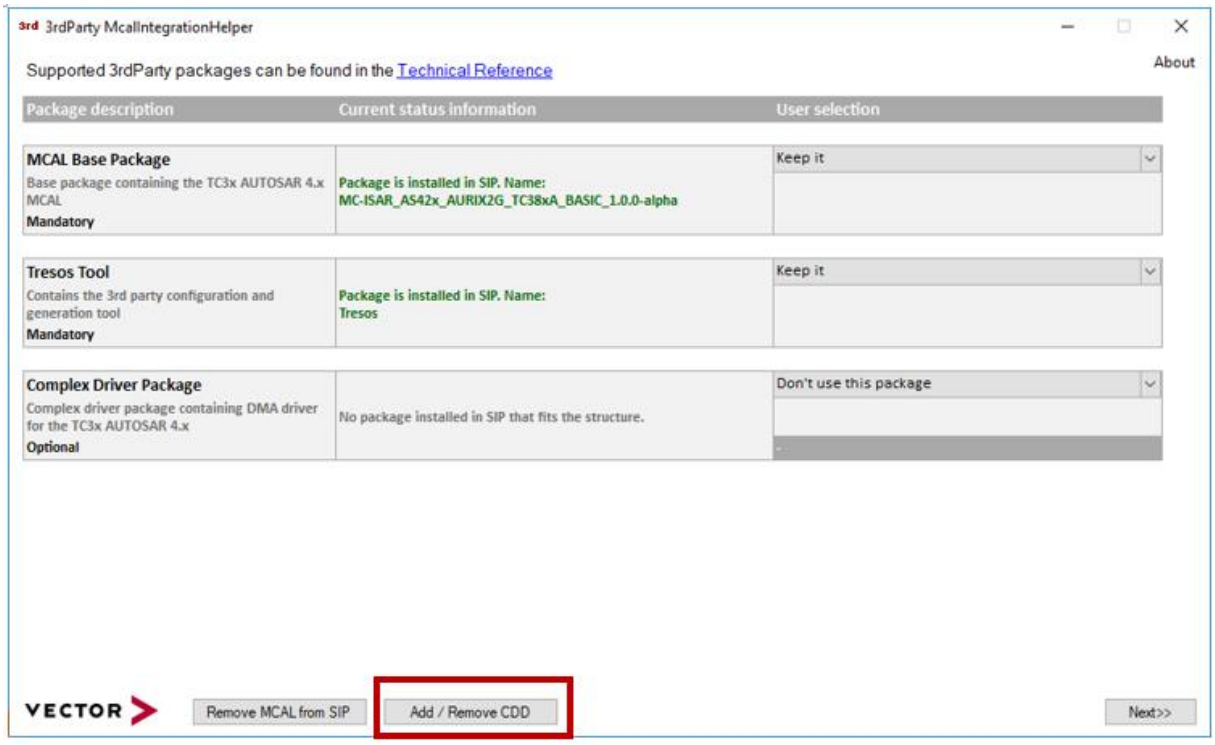


Figure 1-4 Add / Remove CDD button in the integration helper

The “Add / Remove CDD” dialog searches initially for CDD modules in all MCAL packages in `ThirdParty\Mcal_xxx\Supply` folder.

New paths to single CDD modules or to whole CDD packages can also be added:

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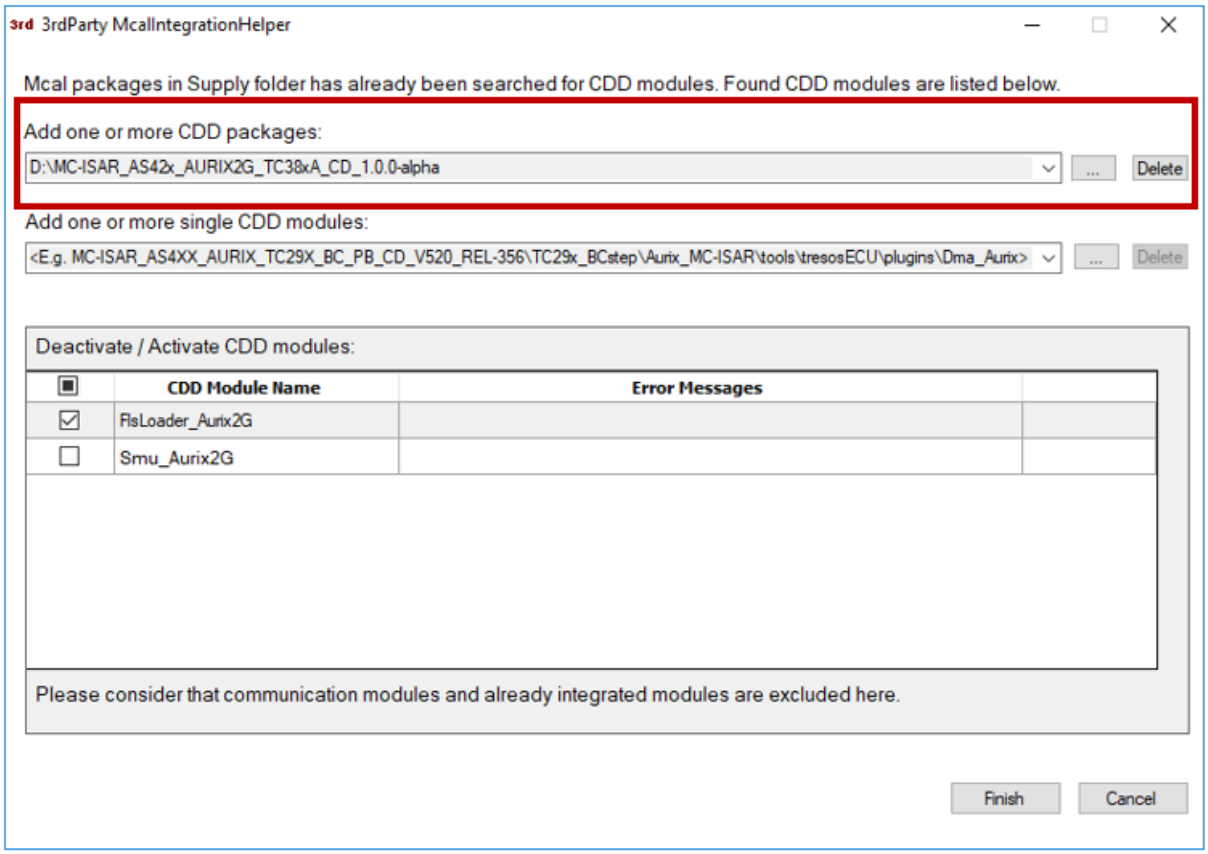


Figure 1-5 Add new path to a CDD package

If no error messages are found for a CDD module (see column called ‘Error Messages’) it can be activated for further integration into the BSW Package:

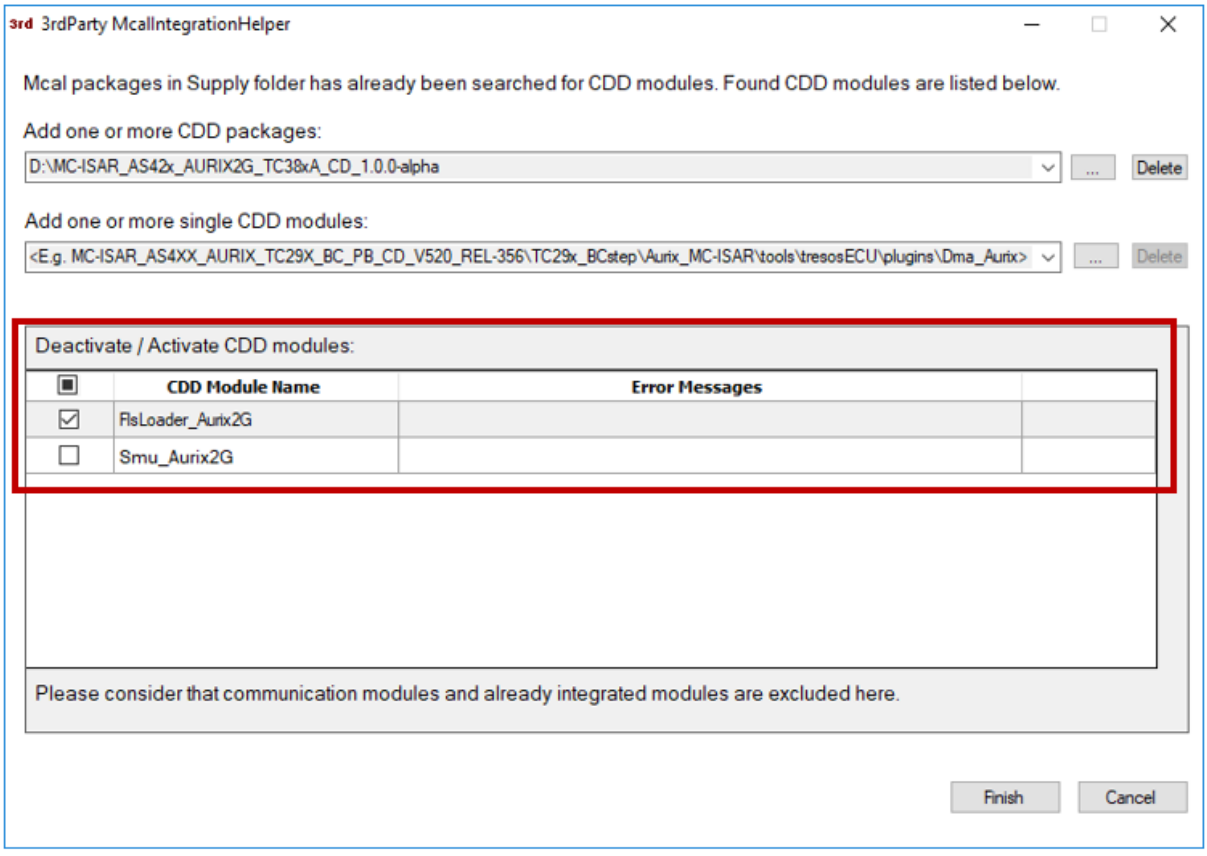


Figure 1-6 De- / Activate CDD modules for integration

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Clicking the ‘Finish’ button does not start immediately the integration of activated CDD modules because the derivative must be selected first. This is done in the further course of the integration helper. Please refer to next chapter.

1.2.1.2 Start integration

The CDD modules are not integrated directly after they have been activated but only after a derivative has been selected. The derivative is selected in the further course of the tool and confirmed with the ‘Copy’ button. The integration result is displayed in a dialog:

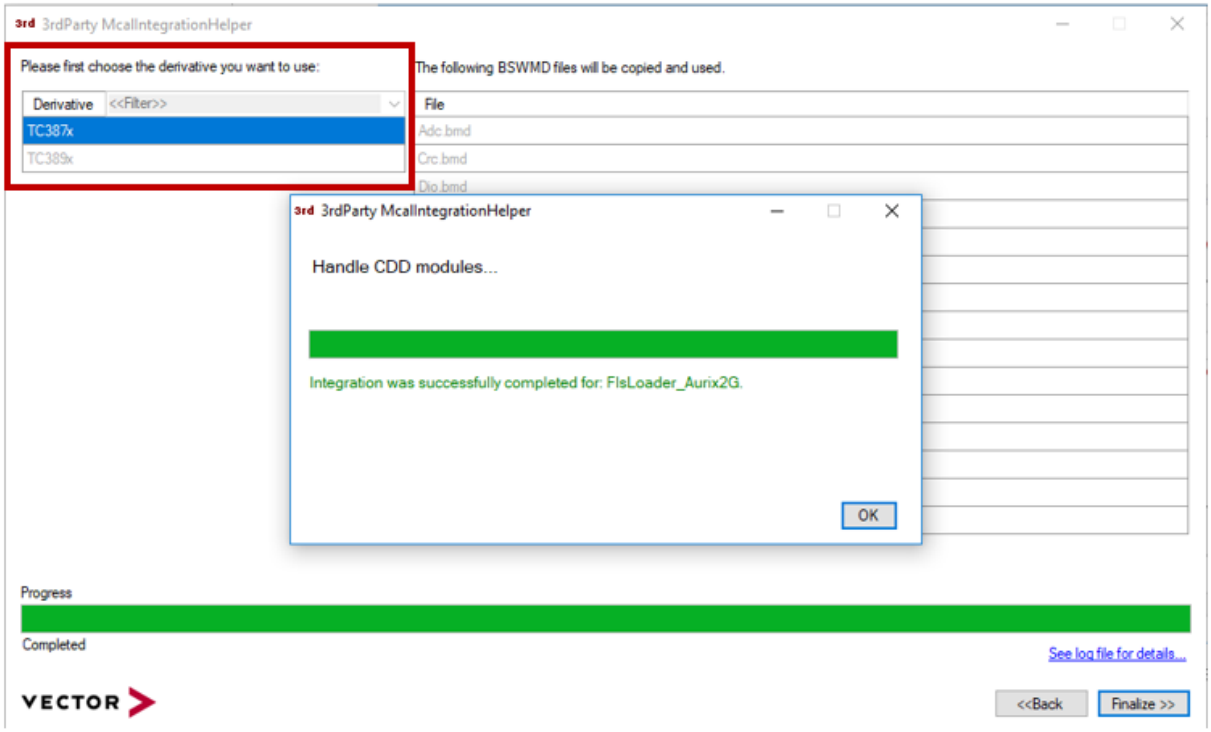


Figure 1-7 CDD integration is executed successfully after derivative was selected

1.2.1.3 Patch integration files

In some cases, the integration can fail. The log file of the integration tool then contains information about where the problem occurred. This can be the case if, for example, dependent modules, which are referenced by a CDD module, are missing. This will cause a generation error in the DaVinci Configurator Classic later.

Therefore, it may be necessary to patch the integration artifacts of a CDD module which can be found in the subfolder of the:

> BSWMD folder of the BSW Package e.g. `BSWMD\<Mcalxxx_CDD_NewModule>` and in the
> Generators folder of the BSW Package e.g. `DaVinciConfigurator\Generators\<Mcalxxx_CDD_NewModule>`

If your BSW Package is available in a component based structure the integration artifacts of a CDD module can be found in the subfolder:

> `Components\<Mcalxxx_CDD_NewModule>`

All these CDD folders, which include the CDD integration artifacts, are newly created every time when the integration helper is running. But before, the last integration artifacts are stored with time stamp in the Temp folder for CDD integrations: `ThirdParty\Mcal_xxx\VectorIntegration\CddHandling\Temp\TimeStamp\<_Mcalxxx_CDD_NewModule>`.

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In this Temp folder you find the last three integrations. Please consider that your patches are overwritten every time integration helper is running. Therefore, you must merge your patched files from the Temp folder with the files in the integration folders after the integration helper run.

The next step is step 5 (chapter 1.5).

1.2.2 Step 2: Integration without CDD feature

1.2.2.4 Copy Tresos plug-in to BSW Package

> Copy the modules, which shall be integrated, into the “plugins” folder of the Tresos instance included in the BSW Package folder `ThirdParty\Mcal_xx\Supply\Tresos\plugins`. This can be the whole content of the “plugins” folder of step 1 `(Tmp\<NewModule>\plugins)` or single modules.

Copy all files (*.jar) and folders to the BSW Package folder, this includes the folder `NewModule_TS_TxxxxxRxx` and the folders of additional referenced modules. The plug-in for the other modules is required because the new module has references to it.



Note
The MCAL is typically not included in the BSW Package after delivery. The customer has to copy his licensed MCAL to the empty third party folder at first by running the integration helper (`3rdPartyMcalIntegrationHelper.exe`).



Reference
Please refer to **TechnicalReference_3rdParty-MCAL-Integration.pdf** for more information about the MCAL integration package provided with the MICROSAR Classic BSW Packages.

If the new module Tresos plug-in requires a different Tresos version than the one in the BSW Package, it is not possible to use one single Tresos instance for generation of all third party modules. Install the correct Tresos version for the new module and copy the plug-ins to this Tresos instance instead of the Tresos instance in the BSW Package as described above. Take care to select the right Tresos instance in the following steps.

The next step is step 3 (chapter 1.3).

1.3 Step 3: Copy BSWMD-rec and BSWMD File

> Create a new folder in the BSWMD resp. Components folder of the BSW Package, e.g.:
legacy structure: `BSWMD\<NewModule>`
component based structure: `Components\<NewModule>\BSWMD`

> Copy one of the existing BSWMD-rec files of the MCAL integration (e.g. `BSWMD\Mcal_xx\`**`Adc`**`_bswmd_rec.arxml` resp. `Components\Mcal_xx\BSWMD\`**`Adc`**`_bswmd_rec.arxml`)
to the new folder `BSWMD\<NewModule>\` resp. `Components\<NewModule>\BSWMD` and rename it to **`<NewModule>`**`_bswmd_rec.arxml`.

> If you do not have an integration package, create **`<NewModule>`**`_bswmd_rec.arxml` files for all modules you like to integrate. An example is shown in Figure 1-10.

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> Copy the actual BSWMD file of the new module from the Tresos “plugins” folder `ThirdParty\Mcal_xx\Supply\Tresos\plugins\NewModule_TS_TxxxxxRxx\autosar\` to the new folder `BSWMD\<NewModule>\` resp. `Components\<NewModule>\BSWMD` and rename it from `<NewModule>.(bmd|epd|arxml)` to **copy**_`<NewModule>.arxml`.

Caution
It is important that the renamed file has **copy_** as prefix.



Figure 1-8 Example BSWMD Folder for New Module Named <NewModule>

Note
In case the MCAL vendor or third party module vendor directly provides arxml files, this last step for coping and renaming the BSWMD is not required and the BSWMD file can be simply put in the corresponding directory.

1.4 Step 4: Adapt BSWMD-rec File

> Open the BSWMD-rec file with an editor and replace the MCAL module name with the new module name at first (e.g. `ADC` to `<NewModule>`).

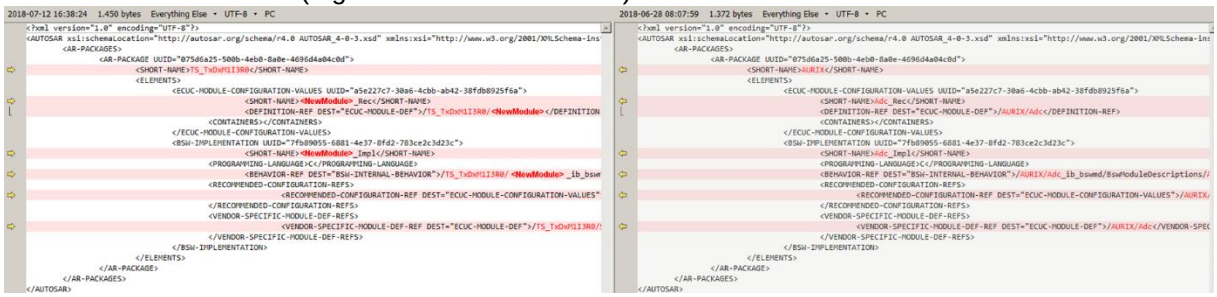
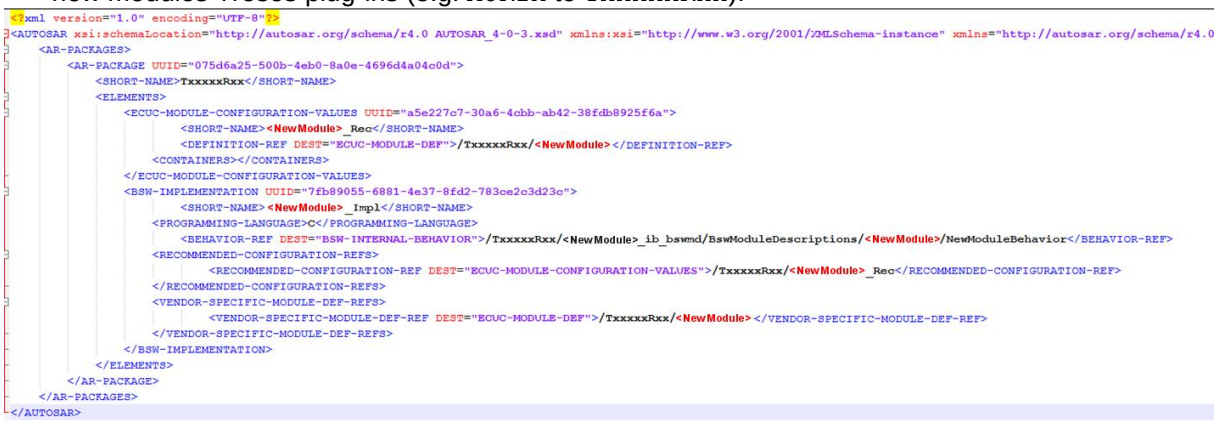


Figure 1-9 Example of Adaptation in <NewModule> BSWMD-rec File Derived from ADC BSWMD-rec File

> Afterwards also replace the short name of the MCAL ADC package with the package name of the new modules Tresos plug-ins (e.g. `AURIX` to `TxxxxxRxx`).



```
<?xml version="1.0" encoding="UTF-8"?>
<AUTOSAR xsi:schemaLocation="http://autosar.org/schema/r4.0 AUTOSAR_4-0-3.xsd" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://autosar.org/schema/r4.0
    <AR-PACKAGES>
        <AR-PACKAGE UUID="075d6a25-500b-4eb0-8a0e-4696d4a04c0d">
            <SHORT-NAME>TxxxxxRxx</SHORT-NAME>
            <ELEMENTS>
                <ECUC-MODULE-CONFIGURATION-VALUES UUID="a5e227c7-30a6-4cbb-ab42-38fdb8925f6a">
                    <SHORT-NAME><NewModule>_Rec</SHORT-NAME>
                    <DEFINITION-REF DEST="ECUC-MODULE-DEF">/TxxxxxRxx/<NewModule></DEFINITION-REF>
                <CONTAINERS></CONTAINERS>
                </ECUC-MODULE-CONFIGURATION-VALUES>
                <BSW-IMPLEMENTATION UUID="7fb89055-6881-4e37-8fd2-783ce2c3d23c">
                    <SHORT-NAME><NewModule>_Impl</SHORT-NAME>
                <PROGRAMMING-LANGUAGE>C</PROGRAMMING-LANGUAGE>
                    <BEHAVIOR-REF DEST="BSW-INTERNAL-BEHAVIOR">/TxxxxxRxx/<NewModule>_ib_bswmd/BswModuleDescriptions/<NewModule>/NewModuleBehavior</BEHAVIOR-REF>
                <RECOMMENDED-CONFIGURATION-REFS>
                    <RECOMMENDED-CONFIGURATION-REF DEST="ECUC-MODULE-CONFIGURATION-VALUES">/TxxxxxRxx/<NewModule>_Rec</RECOMMENDED-CONFIGURATION-REF>
                </RECOMMENDED-CONFIGURATION-REFS>
                <VENDOR-SPECIFIC-MODULE-DEF-REFS>
                    <VENDOR-SPECIFIC-MODULE-DEF-REF DEST="ECUC-MODULE-DEF">/TxxxxxRxx/<NewModule></VENDOR-SPECIFIC-MODULE-DEF-REF>
                </VENDOR-SPECIFIC-MODULE-DEF-REFS>
                </BSW-IMPLEMENTATION>
            </ELEMENTS>
        </AR-PACKAGE>
    </AR-PACKAGES>
</AUTOSAR>
```

Figure 1-10 Example of <New Module> BSWMD-rec File

You can follow with next step.

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1.5 Step 5: Add New Module to ECU Configuration

> Open the DaVinci Configurator Classic and select **Project|Project Settings** in the menu. Click on the blue plus symbol to **Add a new module**.

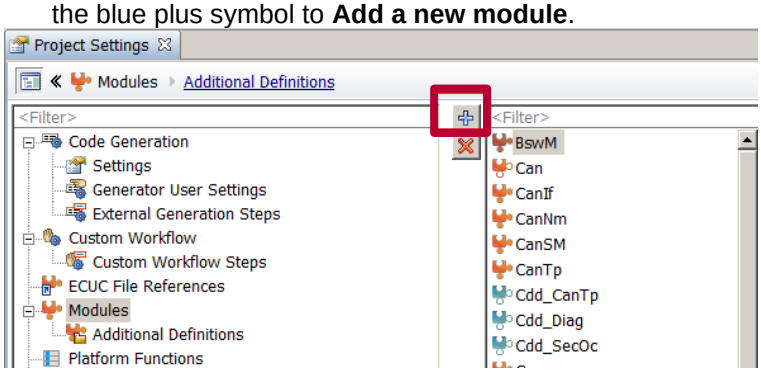


Figure 1-11 Project Settings in DaVinci Configurator Classic

> Click **Select from BSW Package**, because the new module is now recognized as BSW Package module after the BSWMD file has been copied.

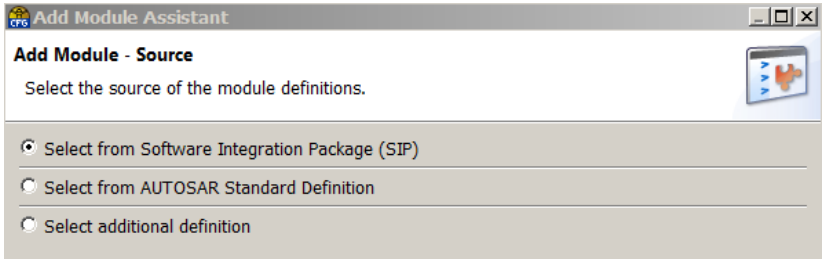


Figure 1-12 Add Module Assistant - Source

> Click the check box to select the new module
> Click **[Finish]**.

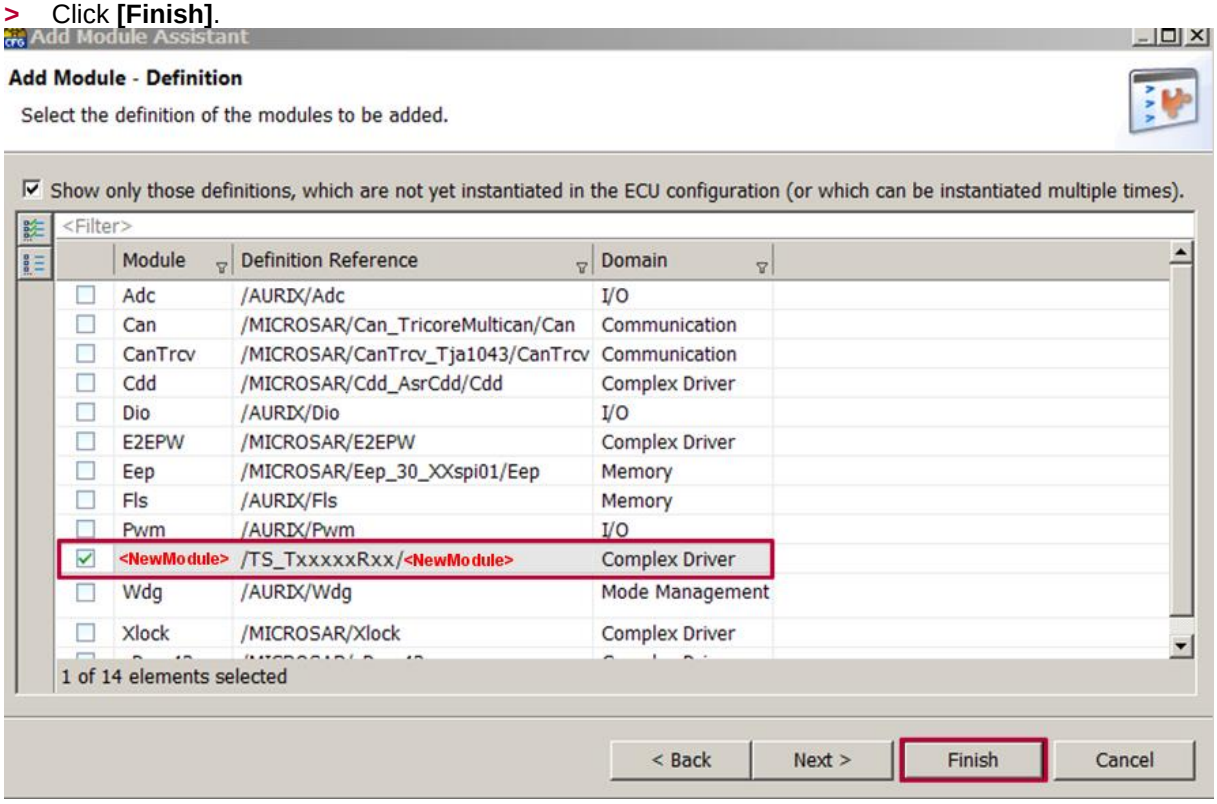


Figure 1-13 Add Module Assistant – Definition

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Continue with next step (step 6).

1.6 Step 6: Configure New Module in Basic Editor

> Open the basic editor and configure the new module like other BSW modules according to the requirements of the OEM or the project.

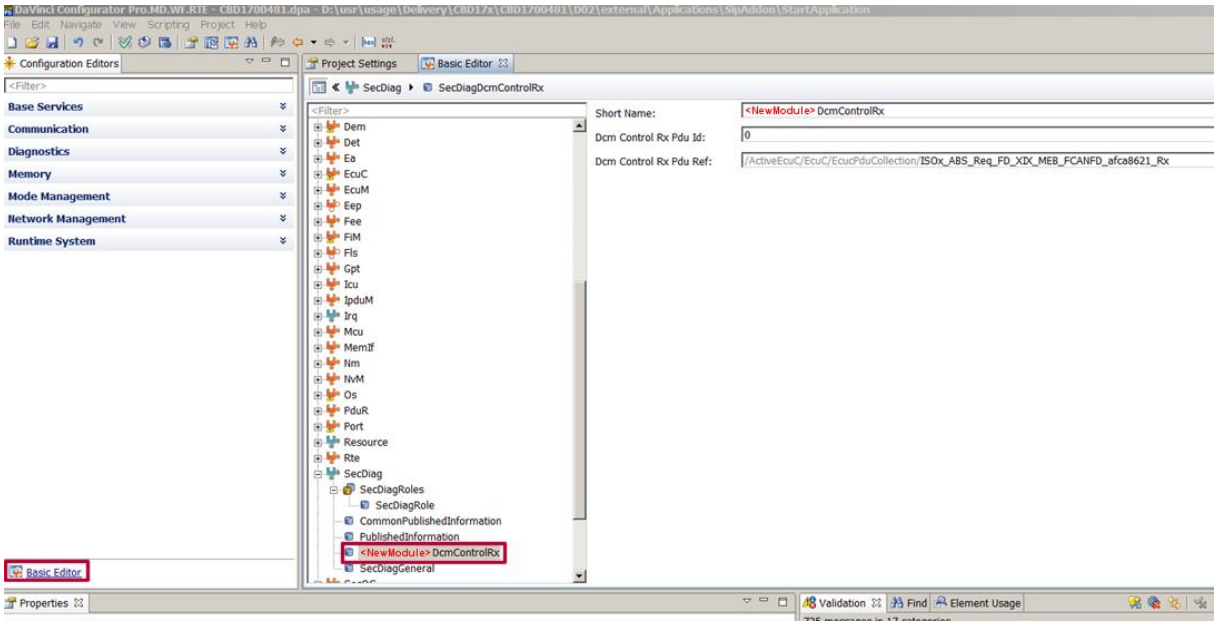


Figure 1-14 Basic Editor DaVinci Configurator Classic

If your module contains exclusive areas, you can either create a special Settings_<NewModule>.xml file (continue with step 7) or you can use the editor “Internal behavior” in DaVinci Configurator:

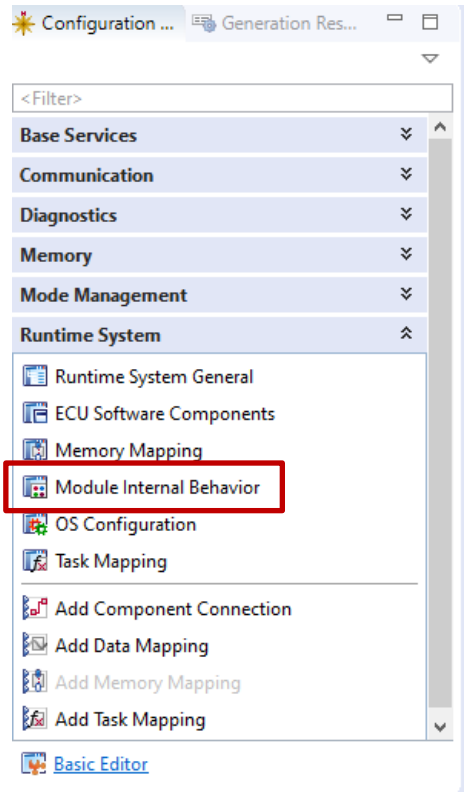


Figure 1-15 Module Internal Behavior Editor DaVinci Configurator Classic

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After finishing the configuration, you start the generation for the new module. To do this, continue with the step 8.

1.7 Step 7: Create Settings_<NewModule>.xml File (Optional)



Note
This step is only required if the new module includes exclusive areas or MainFunctions. Look for `SchM_Enter/Exit` or `MainFunction` in the module code to find out whether this is step is required.

Ff the integration package is used, an example of Settings_<NewModule>.xml file can be found in:

Legacy structure:
`DaVinciConfigurator\Generators\Mcal_xx\Settings_Adc.xml`

Component based structure:

`Components\Mcal_xx\GeneratorMsr\Settings_Adc.xm`

1.7.1 Integration with CDD feature

Like the Tresos settings file described in the chapter 1.2, the `Settings_<NewModules>.xml` file also has to be placed in the folder:
`DaVinciConfigurator\Generators\<Mcalxxx_CDD_NewModule>` in legacy resp.
`Components\<Mcalxxx_CDD_NewModule>\GeneratorMsr` in component based structure.

1.7.2 Integration without CDD feature

Create the folder

`DaVinciConfigurator\Generators\<NewModule>` in legacy resp.
`Components\<NewModule>\GeneratorMsr` in component based structure.

Place the Settings_<NewModule>.xml file into the created folder.

1.7.3 Configuration in RTE

If the Settings file is not automatically recognized by the RTE:

> remove the RTE from the configuration and activate it again.
> After this step you have to reconfigure the RTE (basically perform the auto solving actions and redo the task mapping).

Now you should see the new module inside the container **RteBswModuleInstances**.

1.8 Step 8: Create InitFunctions_Settings.xml file

The file `InitFunctions_Settings.xml` like the shown example below should be created and also put into the folder

`DaVinciConfigurator\Generators\<Mcalxxx_CDD_NewModule>` in legacy resp.
`Components\<Mcalxxx_CDD_NewModule>\GeneratorMsr` in component based structure.

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If the module has no Initialization Function, the InitPhase should be “NO_INIT”. The file can be validated with the schema “InitService.xsd” which is available in DaVinciConfigurator\Core\Schemas.

Example

```
<?xml version="1.0" encoding="UTF-8"?>
<ExtModuleInitialization xmlns="InitService" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="InitService .\InitService.xsd">
    <InitModule>
        <ModuleRef>/TS_TxxxxxRxx/<NewModule></ModuleRef>
    <ModuleInitFunction>
        <InitFunctionName><NewModule>_Init</InitFunctionName>
        <InitFunctionHeader><NewModule>.h</InitFunctionHeader>
        <InitPhase>INIT_ONE_LATE</InitPhase>
    <InitFunctionData>
    <ConfigPtrData>
        <ConfigPtrName><NewModule>_ConfigRoot</ConfigPtrName>
        <ConfigPtrClass>const</ConfigPtrClass>
        <ConfigPtrType><NewModule>_ConfigType</ConfigPtrType>
        </ConfigPtrData>
        </InitFunctionData>
        </ModuleInitFunction>
    </InitModule>
</ExtModuleInitialization>
```

If the file is not created, the Initialization Functions must be configured in DaVinci Configurator Clasic:

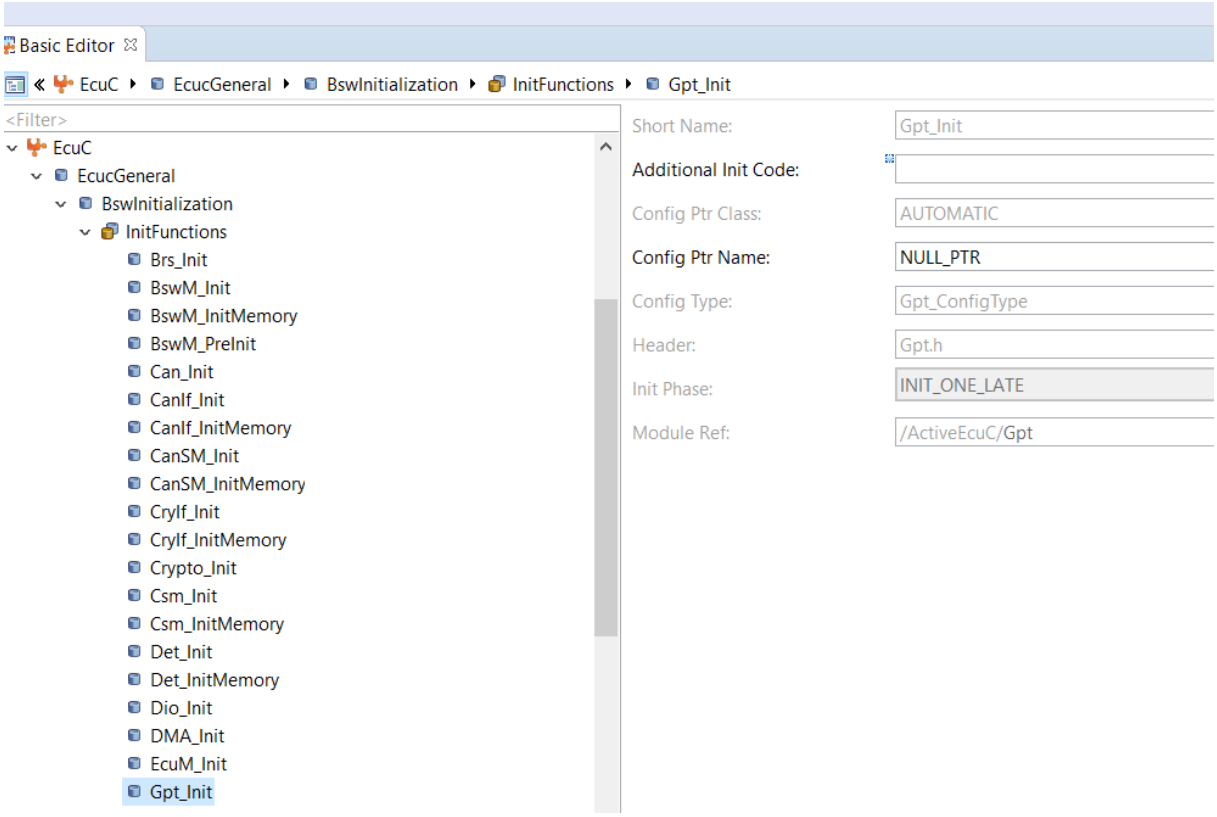


Figure 1-16 Configuration of init functions in EcuC plug-in

In both cases, with and without Init Functions_Settings.xml file, the initialization functions must be activated in the initialization settings:

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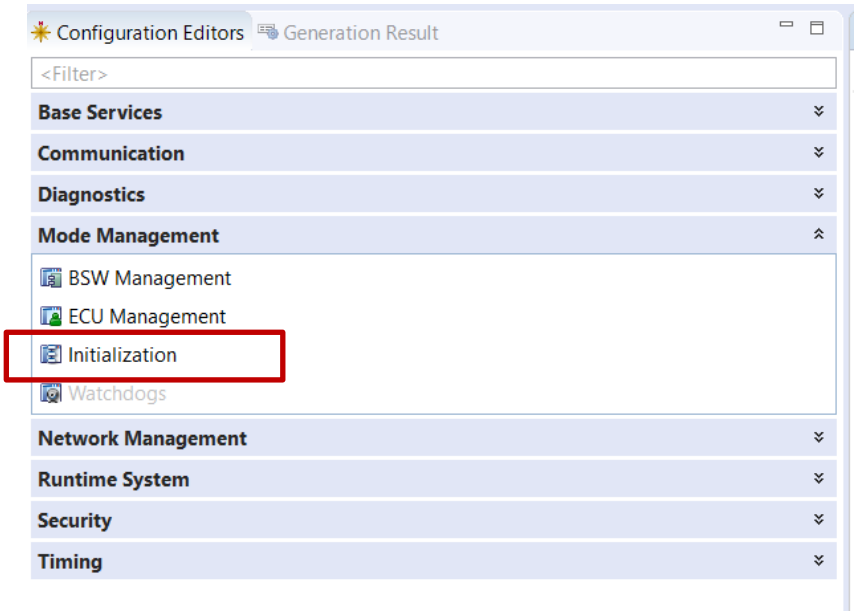


Figure 1-17 Initialization Editor DaVinci Configurator Classic

1.9 Step 9: Start Code Generation

This is only relevant if the integration package including CDD feature is available. Otherwise, please proceed with chapter 1.9.1.

Before you can start the code generation you have to check if the performance optimization of Tresos generator is activated:

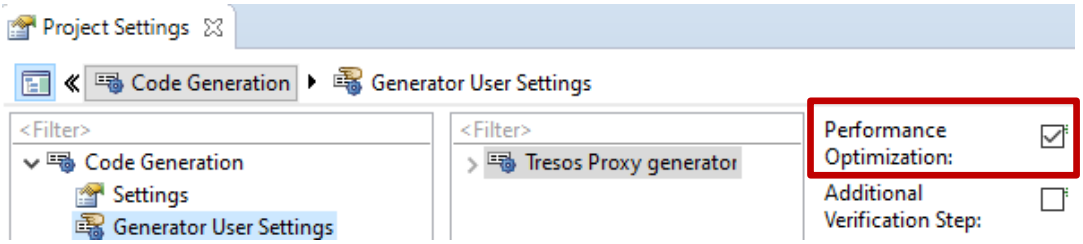


Figure 1-18 Enable performance optimization for Tresos generator

Please note, that the performance optimization ensures that all modules are considered during generation. In some cases, e.g., when an MCAL contains errors, you maybe disable this option to generate single modules instead of all modules. In this case, the CDD integration may fail due to missing modules.

> Now, save the project and start the code generation in DaVinci. Select the new module:

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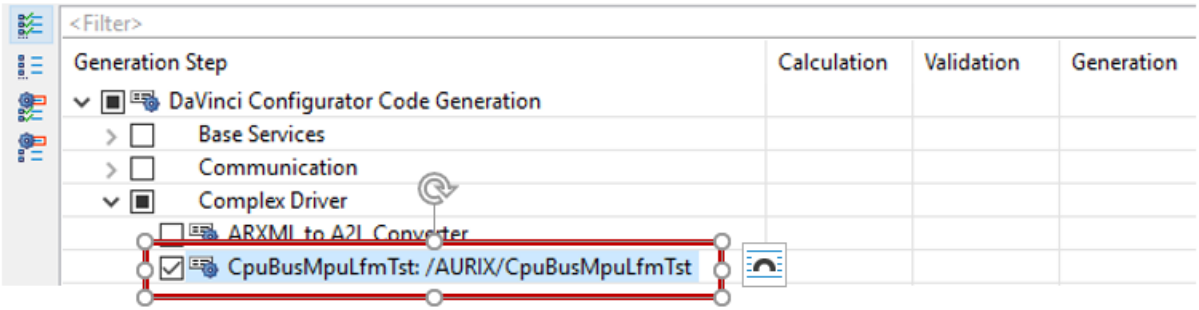


Figure 1-19 DaVinci Configurator Classic Code Generation

The code will be generated to the output path defined in the generated Tresos settings file in step 2. Errors during code generation are reported to the **Generation Result** window in DaVinci Configurator Classic.

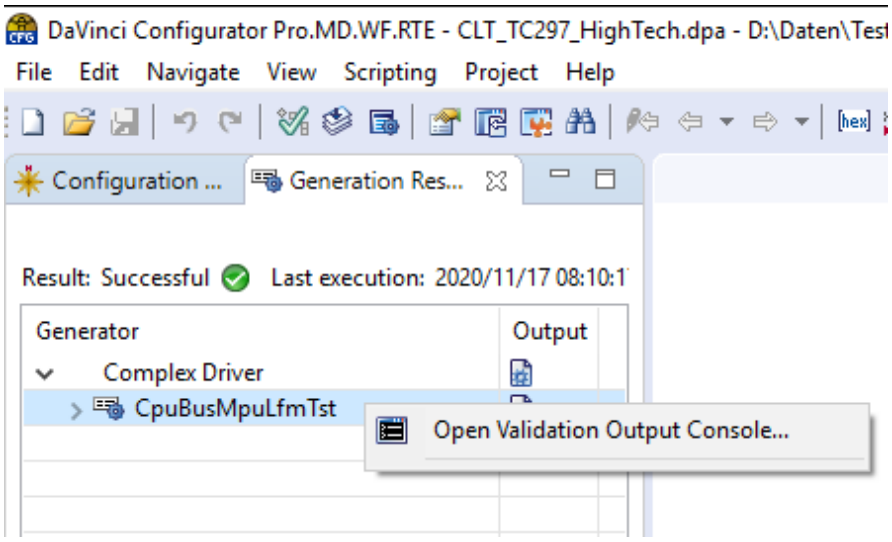


Figure 1-20 Generation Result - Open Generation Output Console…

1.9.1 Add External Generator to DaVinci Project

This step is necessary:

> if other Tresos installation is used than in BSW Package or
> if CDD feature of integration helper was not used or
> if CDD feature was used but other Tresos installation is used than in BSW Package

In all these cases you need to create an external generation step in DaVinci Configurator Classic which starts the right Tresos instance.

> Navigate to **Project|Project Settings|External Generator Steps**
> Click on the blue plus symbol to **Add a new module**.

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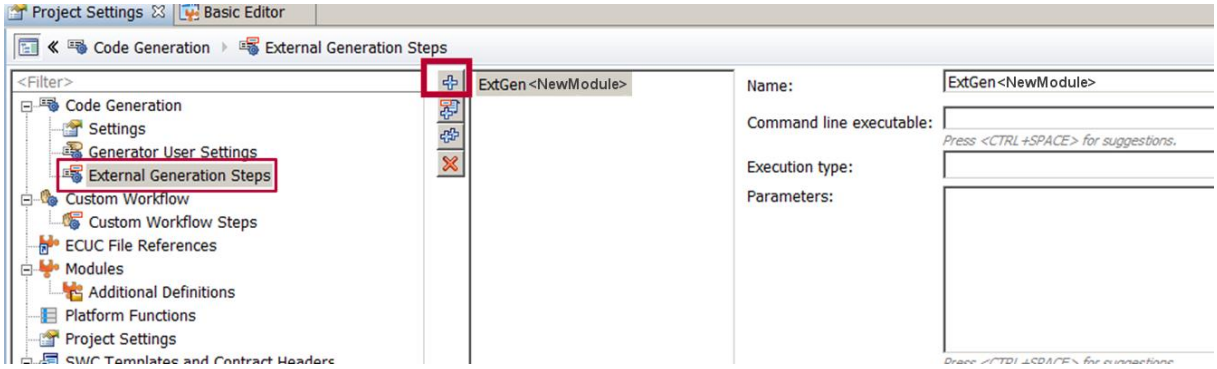


Figure 1-21 External Generation Steps

Fill in the single elements of the new generator dialog:

- **Command line executable**: absolute path to the Tresos instance that shall be used for code generation which can either be the instance in the BSW Package or another one (see step 2) `\ThirdParty\Mcal_xx\Supply\Tresos\bin\tresos_cmd.bat`
- **Execution type**: serial
- **Parameters in legacy structure**: `-Dtarget=AURIX -Dderivate=TC275 legacy generate $(SipRootPath)\DaVinciConfigurator\Generators\NewModule\NewModule_xslt_Result.arxml@asc:4.0.3 -o $(GenDataFolder) -n NewModule -n Module1 -n Module2 -n Module3 -g <NewModule AR-PACKAGE>`

- **Parameters in component based structure**: `-Dtarget=AURIX -Dderivate=TC275 legacy generate $(SipRootPath)\Components\NewModule\GeneratorMsr\NewModule_xslt_Result.arxml@asc:4.0.3 -o $(GenDataFolder) -n NewModule -n Module1 -n Module2 -n Module3 -g <NewModule AR-PACKAGE>`

Reference
Refer to the Tresos command line help in the Tresos GUI for description of the single command line parameters and read the 2.1_Studio_documentation_users_guide.pdf for more information about the legacy generate on the command line.

`NewModule_xslt_Result.arxml` is an intermediate file which can also be placed in a temp folder. All BSW modules, to which the new module has references to (Module1...n), have to be added via the option `-n` and additionally the Resource module, if Resource is part of the MCAL.

- **Working Folder**: `$(SipRootPath)\DaVinciConfigurator\Generators\<NewModule>`
- **AUTOSAR format**: select AUTOSAR format of the new module
- **Transformation file in legacy structure:** `$(SipRootPath)\DaVinciConfigurator\Generators\<NewModule>\prepare_output_NewModule.xsl`
- **Transformation file in component based structure:** `$(SipRootPath)\Components\<NewModule>\GeneratorMsr\prepare_output_NewModule.xsl`

- **Output file in legacy structure:** `$(SipRootPath)\DaVinciConfigurator\Generators\<NewModule>\NewModule_xslt_Result.arxml`
- **Output file in component based structure:** `$(SipRootPath)\Components\<NewModule>\GeneratorMsr\NewModule_xslt_Result.arxml`

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The external generation steps are written to the DPA file after saving the project. One can copy this section to another project DPA file if the third party integration shall be transferred to other projects.

> As an alternative copy this code section in a `Settings_<NewModule>.xml` file and put it in the folder `Davinci\Generators\<NewModule>` resp. `\Components\<NewModule>\GeneratorMsr`

If there already exist external generators for other modules, then it would be the best way to copy the external generators and just adapt the path names in the command line. This is the case if the MCAL or third party module vendor does not use Tresos.

```
<Settings Name="com.vector.cfg.gui.core.generators.ExtGenStepOrder">
    <Setting Value="ExtGenSecDiag|INTERNAL_GENERATION" Name="Order"/>
</Settings>
<Settings Name="com.vector.cfg.gui.core.generators.ExtGenSteps">
    <Settings Name="ExtGenSecDiag">
        <Setting Value="true" Name="Active"/>
        <Setting Value="D:\usr\usage\Delivery\CBD17x\CBD1700481\D02\external\ThirdParty\Mcal_Tc2xx\Supply\Tresos\bin\tresos_cmd.bat" Name=
        <Setting Value="IBM850" Name="CommandLineEncoding"/>
        <Setting Value="" Name="Comment"/>
        <Setting Value="Serial" Name="ExecutionType"/>
        <Setting Value="legacy generate $(SipRootPath)\DaVinciConfigurator\Generators\<NewModule>\<NewModule>_xslt_Result.arxml@asc:4.0.3 -
        <Setting Value="4.0.3" Name="RequiredAsVersion"/>
        <Setting Value="true" Name="SpecificAsVersionRequired"/>
        <Setting Value="false" Name="SupportsStandAloneValidation"/>
        <Setting Value="" Name="TargetType"/>
        <Setting Value="$(SipRootPath)\DaVinciConfigurator\Generators\<NewModule>\<NewModule>_xslt_Result.arxml" Name="TransformationOutput
        <Setting Value="true" Name="TransformationRequired"/>
        <Setting Value="$(SipRootPath)\DaVinciConfigurator\Generators\SecDiag\prepare_output_SecDiag.xsl" Name="TransformationXsltFile"/>
        <Setting Value="" Name="ValidationParameters"/>
        <Setting Value="$(SipRootPath)\DaVinciConfigurator\Generators\<NewModule>" Name="WorkingDir"/>
    </Settings>
</Settings>
```

Figure 1-22 DPA File Snippet to Transfer the New Module Integration to Another Project

> If an external generator was added like described above, select the new module´s external generation step e.g. `ExtGenCpuBusMpuLfmTst` in the Code Generation of DaVinci Configurator Classic:



Figure 1-23 DaVinci Configurator Classic Code Generation: activate external generation step

If it is the case that an external generation step has been created because your module requires a different Tresos version than the one in the BSW Package, then you must switch off the generation created by integration helper´s CDD integration. This ensures that the external generation step is used:

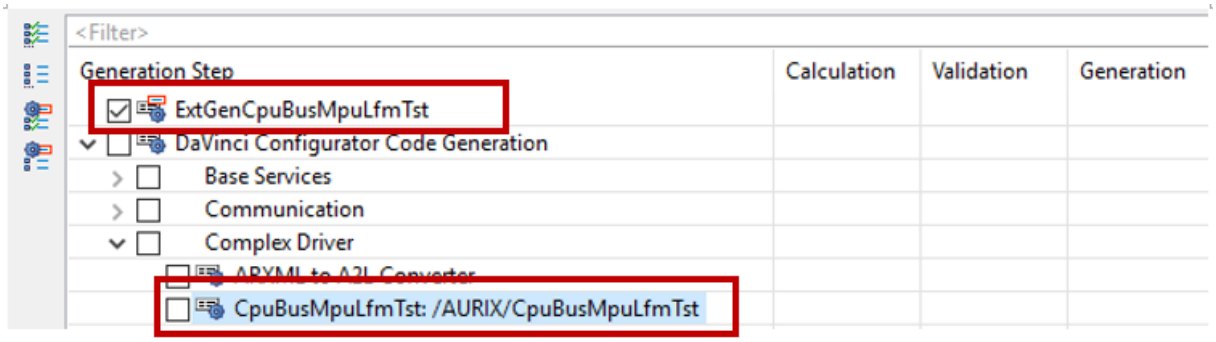


Figure 1-24 DaVinci Configurator Classic Code Generation: activate external generation step & deactivate integrated generation

The code shall be generated to the output path defined above (`-o $(GenDataFolder)`). Errors during code generation with defined external steps are also reported to the **Generation Result** window in DaVinci Configurator Classic.

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2 Additional Resources

USERMANUAL 3RD PARTY MCAL INTEGRATION

UserManual_3rdParty-MCAL-Integration.pdf

3 Contacts

For a full list with all Vector locations and addresses worldwide, please visit http://vector.com/contact/.

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